

Number Sense

Integers · BEDMAS · Exponents · Fractions · Roots · Percent · Ratios

MTH1W · Class — Guided Practice

Name: _____

Date: _____

Goal: Use integers, exponent laws, fractions, roots, percents and ratios to solve multi-step problems.

Warm-up (quick recall)

1. Evaluate: $(-7) + (+3)$.
2. Write $3 \times 3 \times 3$ as a single power, then find its value.
3. Reduce the fraction $20/25$ to lowest terms.

Examples & Practice

For each skill, study the worked **Example**, then solve the questions that follow. Difficulty rises within each set (L1 → L3). Answers are on the last page.

Integers

Example:

Evaluate $(-8) - (-3) + (-5)$.

Add the opposite: $(-8) + (+3) + (-5)$.

Combine: $(-8 + 3) = -5$, then $(-5) + (-5) = -10$. Answer: -10.

4. Evaluate: $(-6) + (-9)$. **[L1]**

5. Evaluate: $(-15) - (-4)$. **[L1]**
6. Evaluate: $(-3) \times 4 + (-2) \times (-5)$. **[L2]**
7. At 6 a.m. the temperature was -4°C . It rose 9°C by noon, then fell 6°C by evening. Find the evening temperature. **[L3]**

Order of Operations (BEDMAS)

Example:

Evaluate $24 - 3 \times (2 + 3)^2 \div 5$.

Brackets: $2 + 3 = 5$. Exponent: $5^2 = 25$.

$3 \times 25 = 75$; $75 \div 5 = 15$; $24 - 15 = 9$. Answer: 9.

8. Evaluate: $8 + 6 \times 2$. **[L1]**

9. Evaluate: $(7 - 2)^2 - 4 \times 3$. **[L2]**

10. Evaluate: $36 \div (2 + 4) + 5 \times 2$. **[L2]**

11. Evaluate: $50 - [3 + 2 \times (4^2 - 10)]$. **[L3]**

Exponent Rules

Example:

Simplify $(x^5 \times x^2) \div x^3$, $x \neq 0$.

Product rule: $x^{5+2} = x^7$. Quotient rule: $x^{7-3} = x^4$.

Answer: x^4 .

12. Write as a single power: $4^3 \times 4^2$. **[L1]**

13. Simplify, $n \neq 0$: $n^6 \div n^2$. **[L1]**

14. Evaluate: $(2^3)^2 \times 2^0$. **[L2]**

15. Explain why $a^3 + a^3 \neq a^6$, and give the correct simplified form. **[L3]**

Fractions — Four Operations

Example:

Evaluate $5/6 - 1/3 \times 3/4$.

Multiply first: $1/3 \times 3/4 = 1/4$.

$5/6 - 1/4$, LCD 12: $10/12 - 3/12 = 7/12$. Answer: $7/12$.

16. Evaluate in lowest terms: $2/5 + 1/10$. **[L1]**

17. Evaluate in lowest terms: $\frac{3}{4} \times \frac{8}{9}$. **[L1]**

18. Evaluate in lowest terms: $\frac{7}{8} \div \frac{3}{4}$. **[L2]**

19. Evaluate in lowest terms: $\frac{1}{2} + \frac{2}{3} \div \frac{4}{3} - \frac{1}{6}$. **[L3]**

Rational Numbers

Example:

Order from least to greatest: -0.75 , $-\frac{2}{3}$, -0.7 , $-\frac{5}{6}$.

As decimals: $-\frac{5}{6} \approx -0.833$, $-\frac{2}{3} \approx -0.667$.

Order: $-\frac{5}{6} < -0.75 < -0.7 < -\frac{2}{3}$.

20. Which is larger: $-\frac{3}{4}$ or $-\frac{2}{3}$? **[L1]**

21. Order from least to greatest: -1.2 , $3/4$, $-1/4$, 1.5 . **[L2]**
22. Find a rational number between $1/3$ and $1/2$. **[L2]**
23. Is the sum of two negative rational numbers always negative? Explain with an example. **[L3]**

Square Roots & Irrationals

Example:

Is $\sqrt{72}$ rational or irrational? Estimate it to one decimal place.

72 is not a perfect square $\rightarrow$ irrational.

$8^2 = 64$ and $9^2 = 81$, so $\sqrt{72} \approx 8.5$.

24. Evaluate: $\sqrt{121}$. **[L1]**

25. Between which two whole numbers does $\sqrt{40}$ lie? **[L1]**

26. Classify each as rational or irrational: $\sqrt{16}$, $\sqrt{17}$, 0.25, π . **[L2]**

27. Explain why $\sqrt{2}$ cannot be written as a terminating or repeating decimal. **[L3]**

Fractions $\leftrightarrow$ Decimals $\leftrightarrow$ Percent

Example:

Convert $\frac{3}{8}$ to a decimal and a percent.

$$3 \div 8 = 0.375.$$

$$0.375 \times 100 = 37.5\%. \text{ Answer: } 0.375 = 37.5\%.$$

28. Write 0.6 as a fraction (lowest terms) and a percent. **[L1]**
29. Write 45% as a fraction in lowest terms. **[L1]**
30. Order from least to greatest: 0.7, $\frac{3}{5}$, 65%. **[L2]**
31. A student scores $\frac{21}{25}$ on a test. Write it as a percent and state whether it is above 80%. **[L3]**

Applying Percents

Example:

A \$90 item is discounted 30%. Find the sale price.

Discount = $0.30 \times 90 = \$27$.

Sale price = $90 - 27 = \$63$ (or $90 \times 0.70 = \$63$).

32. Find 15% of 200. **[L1]**

33. A \$45 shirt has 13% HST added. Find the total cost. **[L2]**

34. A population grows from 400 to 460. Find the percent increase. **[L2]**

35. After a 20% discount a game costs \$72. Work backwards to find the original price.
[L3]

Ratios, Rates & Proportions

Example:

4 pens cost \$5.00. Find the cost of 10 pens.

Unit rate: $5.00 \div 4 = \$1.25$ per pen.

$10 \times 1.25 = \$12.50$. Answer: \$12.50.

36. Simplify the ratio 18 : 24. **[L1]**
37. A car uses 6 L of fuel per 100 km. How much fuel is needed for 250 km? **[L2]**
38. Share \$60 in the ratio 2 : 3. **[L2]**
39. A recipe for 4 servings uses 300 g of flour. How much flour is needed for 10 servings? **[L3]**

Answers

Warm-up: W1) -4 W2) $3^3 = 27$ W3) $4/5$

4. (L1) -15
5. (L1) -11
6. (L2) -2
7. (L3) $-4 + 9 - 6 = -1$ °C
8. (L1) 20
9. (L2) 13
10. (L2) 16
11. (L3) $4^2 - 10 = 6$; $2 \times 6 = 12$; $3 + 12 = 15$; $50 - 15 = 35$
12. (L1) 4^5
13. (L1) n^4
14. (L2) $2^6 \times 1 = 64$
15. (L3) Adding like terms adds coefficients, not exponents: $a^3 + a^3 = 2a^3$.
16. (L1) $1/2$
17. (L1) $2/3$
18. (L2) $7/6$
19. (L3) $\div$ first: $2/3 \div 4/3 = 1/2$; then $1/2 + 1/2 - 1/6 = 5/6$
20. (L1) $-2/3$ (since $-0.667 > -0.75$)
21. (L2) -1.2, $-1/4$, $3/4$, 1.5
22. (L2) e.g. $5/12$ (the average of $1/3$ and $1/2$)
23. (L3) Yes. e.g. $-1/2 + (-1/3) = -5/6 < 0$; adding two negatives always gives a negative.
24. (L1) 11 (rational)
25. (L1) 6 and 7 ($36 < 40 < 49$)
26. (L2) $\sqrt{16}$ rational, $\sqrt{17}$ irrational, 0.25 rational, π irrational
27. (L3) $\sqrt{2}$ is irrational — it is not a ratio of integers, so its decimal never terminates or repeats.
28. (L1) $3/5$; 60%

- 29.** (L1) $9/20$
- 30.** (L2) $3/5$ (0.6), 65% (0.65), 0.7
- 31.** (L3) $21/25 = 84\%$; yes, $84\% > 80\%$.
- 32.** (L1) 30
- 33.** (L2) $45 \times 1.13 = \$50.85$
- 34.** (L2) $60 \div 400 = 15\%$
- 35.** (L3) $0.80 \times p = 72 \rightarrow p = 72 \div 0.80 = \90
- 36.** (L1) 3 : 4
- 37.** (L2) $6 \times 2.5 = 15$ L
- 38.** (L2) \$24 and \$36
- 39.** (L3) $300 \div 4 \times 10 = 750$ g